Using `TextMidiFormEdit`

The `textmidicgm` form editor

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1 Overview

The TextMidiFormEdit program can be used interactively to edit new or pre-existing musical form files used by textmidicgm, which generates works of music in textmidi language, which can then be converted by textmidi into standard MIDI 1.1 file formats. Using TextMidiFormEdit, it is possible to edit musical forms as defined by textmidicgm rather than hand-edit the XML form file.

The model of musical form defined by textmidicgm is described in the PDF or info files for textmidicgm. This manual only describes the user interface for TextMidiFormEdit.py.

1.1 What is TextMidiFormEdit?

TextMidiFormEdit is a graphical program for editing form files to be used by textmidicgm to create random (random as in windchimes) music. TextMidiFormEdit can be used interactively to create new musical form files, or edit previously-existing musical form files. The program textmidicgm is not an "AI" application and does not train on other music.

2 Downloading TextMidiFormEdit

If you have downloaded `TextMidiFormEdit.py` and its support *.py files, then you can run it as described below.

3 Invoking TextMidiFormEdit

`TextMidiFormEdit.py` can be run either without options or with one option: the filename of a pre-existing form XML file. To run it, go to the directory where it is found (after unpacking the archive for TextMIDITools) and invoke it:

```
./TextMidiFormEdit.py
```

You may specify an existing form file to be edited on the command line:

```
./TextMidiFormEdit.py symphony.xml
```

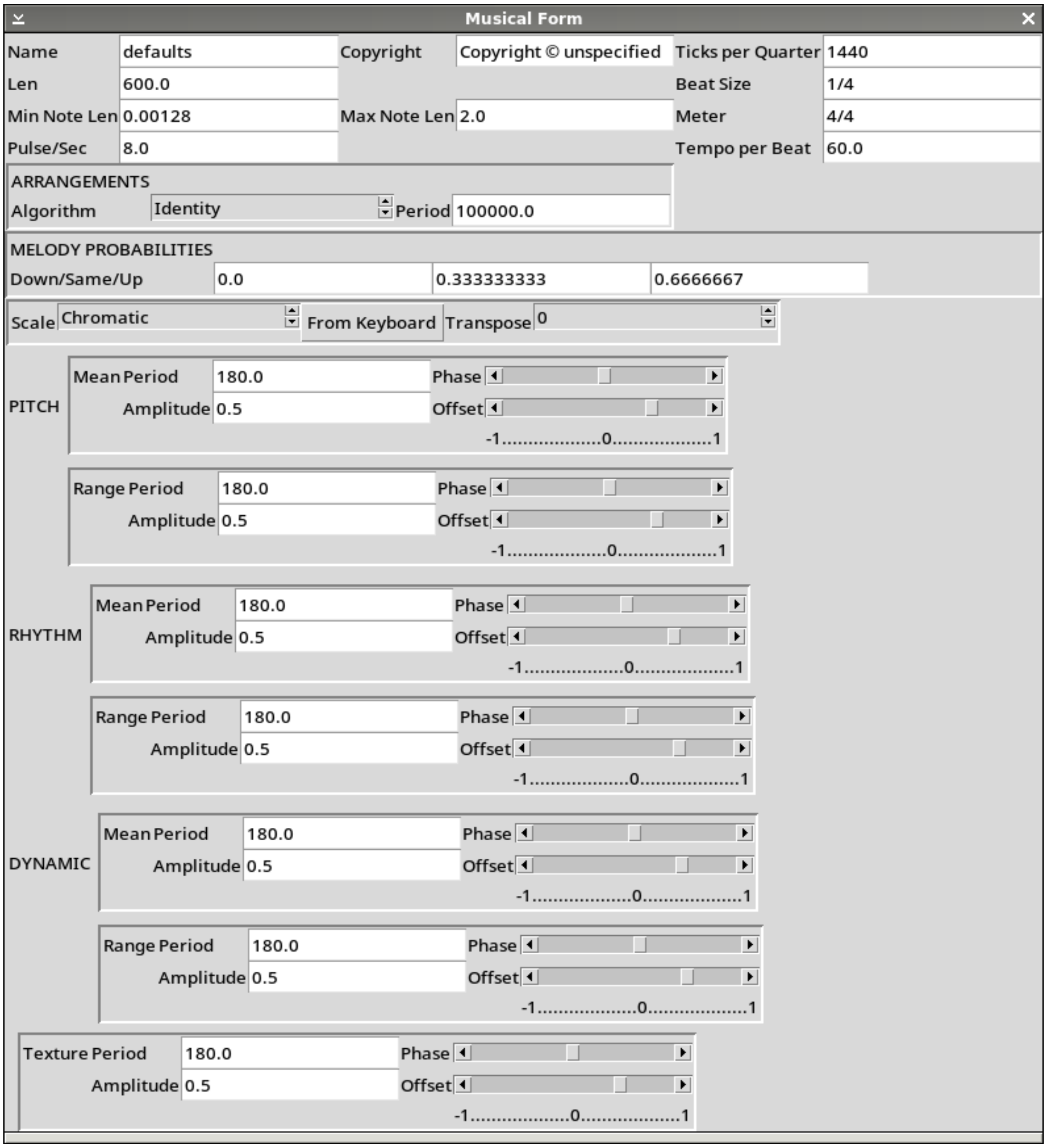
or open a form file using the menus. TextMidiFormEdit can also be run in a more step-wise manner as a Python class, but a filename is required.

```
python3
>>> from TextMidiFormEdit import *
>>> TextMidiFormEdit("filename.xml")
```

When you quit the application, just exit from python:

```
>>> exit()
```

4 Musical Form Window



The Musical Form window allows the setting of global form qualities except for the instrumental attributes of individual voices, which is done on the Voices window. The Musical Form Window has several items.

4.1 Name

The name string can be typed in; it is meant to be the title of the musical work to be created by `textmidicgm`. The name string is offered as the basename of all file save names but the files may be saved to any name.

4.2 Copyright

The copyright string can be typed in. It is not copied to the MIDI file that is created from the form file by `textmidicgm`.

4.3 Len

The len text field is the duration of the musical work in seconds. Entering 600 will tell `textmidicgm` to generate 10 minutes of music.

4.4 Min/Max Note Len

The minimum and maximum note lengths (given in seconds as floating-point numbers) specify the duration range for all note events. When the rhythm curve is very low and the range curve narrow, the min note length is used; when the rhythm curve is high with a narrow range, the maximum note length is used. The original `AlgoRhythms` for the Amiga computer was a real-time program that just pushed note events out the MIDI port as fast as it could, so that the minimum was the transmission time of a note-on/note-off pair; a minimum note length of 0.0 seconds never actually arose (but still was fast enough to make my piano synth hang up). Because `textmidicgm` is merely writing a textmidi-format file, it can try to use 0-long notes, but then musical time would not advance. This is because time advances by adding the note duration to the time (from the start of the piece) of the last note. If time does not advance, then the trigger for ending the piece will not occur. Therefore, huge numbers of notes can be generated, resulting in huge files. That is why there is a limit on the number of notes (which can be changed on the command line). See the documentation for `textmidicgm`.

4.5 Pulse/Sec

Pulse per second is used optionally to force `textmidicgm` to generate rhythms that line up on a pulse. Such music is likely to be easier to notate in scoring program such that human players could play it. Note, however, that `textmidicgm` has no model of meter, that is, the grouping of beats and accents. The original concept for this model of form (developed in 1976 for manual composition) was connected to very fluid, conversational, ametrical rhythms. On the other hand, setting a pulse can make works produced by `textmidicgm` sound more dance-like, for example, with a pulse of about 5 or so per second. A pulse-per-second that is less than 1.0 is allowed. For example, a value of 0.5 pulses per second would be one pulse in 2 seconds. Setting a pulse-per-second around 8.0 or slower is recommended for those who wish to make human-playable scores using `textmidicgm`.

4.6 Ticks per Quarter

MIDI (1) requires specifying ticks per quarter [note], and MIDI (1) tempo is always for quarter note; it does, however allow giving a meter in a MIDI file. `textmidicgm` permits specifying the size of the beat to other than a quarter note, the tempo for that size of beat, and of course the meter. The musical time values of ticks/quarter, beat size, meter and tempo per beat are conveniences that specify how the `textmidi` language file is to be written. They do not, in principle, add to the capability of `textmidicgm`. Ticks per quarter could be as low as 24, is often 240, but can be a higher value such as 720 or even 1440. It may be helpful to have a factor-rich value for ticks/quarter so that divisions such as triplets, quintuplets, and the usual divisions of 2 can be exact. For example, 720 is the product of 2*2*2*2*3*3*5. A quarter note is already 1/(2*2) of a whole note, so that 720 ticks per quarter can exact represent 64th notes as well as 1/8 triplets and quintuplets down to 1/16ths. A value of ticks per quarter that results in higher resolution that your MIDI playing software's timing offers is probably not meaningful.

4.7 Beat Size

The beat size is the musical ratio for the size of a beat. In 6/8 meter, for example, you could specify a beat size of 3/8. The textmidi language *TEMPO* now allows specification of tempo for any size beat, but by default applies to a quarter note.

4.8 Meter

meter is written as you give it, but with separate number of beats and beat size, into the `textmidi` file as a TIME_SIGNATURE command, along with ticks per quarter.

4.9 Tempo per Beat

The tempo per beat is the metronomic tempo of beats per minute for the beat size specified in the *beat size* field. It is converted into a tempo of quarters per minute for the `textmidicgm` language output because that is how MIDI (1) uses tempo. `textmidicgm` will compute rhythms based on beat size, meter, and tempo per beat. It can now write the TEMPO statement as a tempo for a special-size beat because the textmidi language's TEMPO statement has been made to accept a beat size. The program `textmidi` will convert that TEMPO command to the MIDI (1) version of TEMPO for a quarter note. For example, a meter of 60 for a beat size of 3/8 and with 1440 ticks/quarter will be written as a command TEMPO 90 1440.

4.10 Arrangements

Arrangements, if set, can change the priority of voices under the texture curve. Each arrangement algorithm permutes the voice priorities in a different way. Some of the algorithms do not produce all possible permutations, but others do. These algorithms are described in the documentation for `textmidicgm`. The settable items for Arrangements are:

- Algorithm
  - Undefined
  - Identity
  - LexicographicForward
  - LexicographicBackward
  - RotateRight
  - RotateLeft
  - Reverse
  - SwapPairs
  - Skip
  - Shuffle
  - Heaps
- Period The period of time alotted to each permutation before changing to the next one.

4.11 Melodic Probabilities

Melodic Probabilities are cumulative probabilities that apply only when a voice is in walking mode. The probabilities act as thresholds. The thresholds determine how often a voice moves up, or moves down, repeats a note, or is resting. A random variable used to select melodic movement varies in (0..<1.0). When the random variable is:

- Below the "Down" probability, the voice will move down a step in the scale; when ;
- At or above Down and below the Same probability, it will repeat the previous key;
- At or above the Up probability, it will step up the scale.
- At or above the Up probability, it will be silent.

Because these are cumulative probabilities, the following inequalities must hold: 0.0 <= Down <= Same <= Up <= 1.0. If you don't want the voice to drift off down or up, then the probabilities for up and down should be equal, but by differences: (Up - Same) == (Same - Down). These probabilities apply only if a voice is in walking mode.

4.12 Scale

Scales can be entered the following ways:

- Loaded from a form file If you hand-edit the scale inside the form file, remember to change the number inside the tag <*count*> to the number of pitches in the scale. XML format is a bit fragile, and boost::serialization XML archive format is even more fragile.

- Set from the scale menu selections Select a scale by name in the Scale spin button. The scale will be reflected on the Keyboard window.
- Loaded from the keyboard window, by using the "From Keyboard" push button. There is more information about the Keyboard window below.

Once a scale has been loaded, it may be transposed by using the transpose spin button. Note that if the scale loaded from a file was already transposed, then that version of the scale will be the transposition of zero (0) in the Musical Form window for that scale. Transposition will be reflected in the Keyboard window.

4.13 Pitch Form

The Form curves are the core concept of the `textmidicgm` model for musical form. The curves are always sinusoidal (wave-like), but they are usually biased to sit above zero and range from 0 to 1. Pitch, Rhythm, and Dynamic curves have two sinusoids: mean and range. The controls for each sinusoid include a period (in seconds) and a phase slider in radians. There are two * pi radians (about 6.28) in a full circle. The slider ranges from -pi to +pi. For example, to get 90 degrees, set the slider about 3/4 of the way across between 0 and +pi. The mean sinusoid is roughly the average or center value of the parameter (in which parameter refers to pitch, rhythm, and dynamic) and the sinusoid for range gives the range from low value to high value. Only a range sinusoid is used for texture. The seven sinusoids are all all independent in phase and period. After adjusting form curve values, you should select the Redraw item on the Form Plot window's menu to see how you have changed the form. The result of these curves is to cause the music generated by `textmidicgm` to change in character over time. Ordinarily the periods used are around 120 to 360 seconds (2 to 6 minutes). However, to make a parameter's mean or range stay static, merely make the period very long, such as 1000000 seconds. Unless a voice is in walking mode, the Pitch mean curve generally causes the pitches played to be higher when the curve is higher and lower when the curve is lowe; the Pitch range curve causes a wider range of of notes in the scale when the curve is higher and fewer (even just repeating a pitch). In walking mode, a voice ignores the pitch curves and follows the melodic probabilities on how to walk up and down. Adjusting amplitude and offset can help avoid the voice spending too long playing extremely high or low notes (or loud or soft, or long or short). When I manually wrote the first piece using this model, in 1977, I appear to have subjectively adjusted, perhaps in a logarithmic-like compression of the curve as it exceeded the limits. This type of adjustment is, however, not available in `textmidicgm`.

4.14 Rhythm Form

A high rhythm mean curve generally tells `textmidicgm` to make longer duration (slower) note events. When the rhythm mean curve is at a low point in the wave, it tends to make shorter durations (faster notes). A high rhythm range curve tends to make a wider range of durations and sound more syncopated, even with a pulse-per-second set; a low rhythm range curve tends to make more monotonous and repetitive durations.

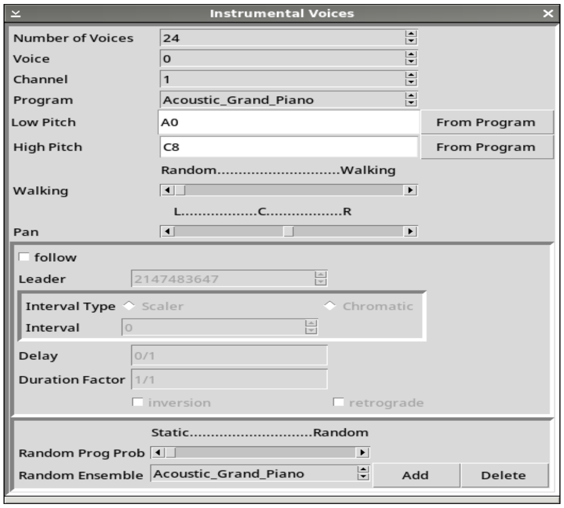
4.15 Dynamic Form

A high dynamic mean curve generally tells `textmidicgm` to make louder note events. When the dynamic mean curve is at a low point in the wave, it tends to make quieter notes. A high dynamic range curve tends to make a wider range of loudnesses and sound more natural; a low dynamic range curve tends to make more evenly loud notes.

4.16 Texture Form

A high texture range curve makes more voices playe (a larger complement); a low texture range curve makes fewer voices play. As the texture curve rises, the voices are added in voice number order. To suggest a solist, the solo voice should probably be in voice 0 position in the voice window.

5 Voices Window



The Voices Window has several items.

5.1 Number of Voices

The number of voices is the total number of voices that can play when the form's texture curve is high. This model of musical form is basically counterpuntal and made of solo voices. There is no limit for number of voices, but the form editor only offers 128 voices.

5.2 Voice

Voice is the index (in 0..Number of Voices - 1) of the voice to be currently edited in the voice window.

5.3 Low Pitch

The low pitch is the low pitch for the voice. If the Form Pitch curve goes below this low pitch, then the low pitch will be played. This can be monotonous; `textmidicgm` has an option on the command that trims the scale to the range for all the instruments together to help with this, but this is not a perfect fix. The "From Program" push button will set the Low Pitch suitable for the patch. The instrument ranges were taken from textbooks or from en.wikipedia.org.

Patch	**Low Pitch**	**High Pitch**
Acoustic_Grand_Piano	A0	C8
Bright_Acoustic_Piano	A0	C8
Electric_Grand_Piano	E1	E7
Honky-tonk_Piano	A0	C8
Electric_Piano_1(Fender_Rhodes)	A0	C8
Electric_Piano_2(DX-7_EP)	A0	C8
Harpsichord	F1	F6
Clavi	F1	F6

Chromatic Percussion		
Celesta	C4	C8
Glockenspiel	F5	C8
Music_Box	F6	C8
Vibraphone	F3	F6
Marimba	C2	C7
Xylophone	C3	C7
Tubular	15	C4 G5
Dulcimer	C3	C5
Organ		
Drawbar_Organ	A0	C8
Percussive_Organ	A0	C8
Rock_Organ	A0	C8
Church_Organ	A0	C8
Reed_Organ	C2	C6
Accordion	E1	D6
Harmonica	C3	G5
Tango	24	E1 D6
Guitar		
Acoustic_Guitar_(nylon)	E3	D6
Acoustic_Guitar_(steel)	E3	D6
Electric_Guitar_(jazz)	E3	D6
Electric_Guitar_(clean)	E3	D6
Electric_Guitar_(muted)	E3	D6
Overdriven_Guitar	E3	D6
Distortion_Guitar	E3	D6
Guitar_Harmonics	E4	D7
Bass		
Acoustic_Bass	E1	E4
Electric_Bass_(finger)	E1	E4
Electric_Bass_(pick)	E1	E4
Fretless_Bass	E1	E4
Slap_Bass_1	E1	E4
Slap_Bass_2	E1	E4
Synth_Bass_1	E1	E4
Synth_Bass_2	E1	E4
Strings		
Violin	G3	C7
Viola	C3	D7
Cello	C2	A5
Contrabass	E1	G4
Tremolo_Strings	E1	C7
Pizzicato_Strings	E1	C7
Orchestral_Harp	B0	G#7
Timpani	C2	D4
Ensemble		
String_Ensemble_1	C0	C8

String_Ensemble_2	C0	C8
Synth_Strings_1	C0	C8
Synth_Strings_2	C0	C8
Choir_Aahs	D2	C6
Voice_Oohs	D2	C6
Synth_Voice	D2	C6
Orchestra_Hit	C0	C8
Brass		
Trumpet	F#3	C6
Trombone	C2	C5
Tuba	Eb2	A4
Muted_Trumpet	F#3	C6
French_Horn	C2	C6
Brass_Section	C2	C6
Synth_Brass_1	C2	C6
Synth_Brass_2	C2	C6
Reed		
Soprano_Sax	Ab3	Eb6
Alto_Sax	Db3	Ab5
Tenor_Sax	Ab2	Eb5
Baritone_Sax	Db2	Ab4
Oboe	Bb3	G6
English_Horn	E3	A5
Bassoon	Bb1	E5
Clarinet	D3	Bb6
Pipe		
Piccolo	D5	C8
Flute	C4	C7
Recorder	C-1	G9
Pan_Flute	C-1	G9
Blown_bottle	C-1	G9
Shakuhachi	C-1	G9
Whistle	C-1	G9
Ocarina	C-1	G9
Synth Lead		
Lead_1_(square)	C0	C8
Lead_2_(sawtooth)	C0	C8
Lead_3_(calliope)	C0	C8
Lead_4_(chiff)	C0	C8
Lead_5_(charang)	C0	C8
Lead_6_(voice)	C0	C8
Lead_7_(fifths)	C0	C8
Lead_8_(bass_+_lead)	C0	C8
Synth Pad		
Pad_1_(new_age)	C0	C8
Pad_2_(warm)	C0	C8
Pad_3_(polysynth)	C0	C8

Pad_4_(choir)	C2	G5
Pad_5_(bowed)	C0	C8
Pad_6_(metallic)	C0	C8
Pad_7_(halo)	C0	C8
Pad_8_(sweep)	C0	C8
Synth Effects		
FX_1_(rain)	C0	C8
FX_2_(soundtrack)	C0	C8
FX_3_(crystal)	C0	C8
FX_4_(atmosphere)	C0	C8
FX_5_(brightness)	C0	C8
FX_6_(goblins)	C0	C8
FX_7_(echoes)	C0	C8
FX_8_(sci-fi)	C0	C8
Ethnic		
Sitar	C2	C6
Banjo	D3	C6
Shamisen	E3	C6
Koto	C2	C6
Kalimba	G3	G6
Bag	110	C3 C6
Fiddle	G3	C7
Shanai	Bb3	G6
Percussive		
Tinkle	C-1	G9
Agogô	C-1	G9
Steel_Drums	C-1	G9
Woodblock	C-1	G9
Taiko_Drum	C-1	G9
Melodic_Tom	C-1	G9
Synth_Drum	C-1	G9
Reverse_Cymbal	C-1	G9
Sound Effects		
Guitar_Fret_Noise	C-1	G9
Breath_Noise	C-1	G9
Seashore	C-1	G9
Bird_Tweet	C-1	G9
Telephone_Ring	C-1	G9
Helicopter	C-1	G9
Applause	C-1	G9
Gunshot	C-1	G9

5.4 High Pitch

The high pitch is the high pitch for the voice. If the Form Pitch curve goes above this high pitch, then the high pitch will be played. This can be monotonous; `textmidicgm` has an option on the command that trims the scale to the range for all the instruments together to help with this, but this is not a perfect fix. The “From Program” push button will set the High Pitch suitable for the patch.

5.5 Channel

The channel is the MIDI Channel (1..16) for the voice.

5.6 Program

The Program is the MIDI program name from General MIDI used for the current voice.

5.7 Walking

The walking value sets the probability of a voice being in walking mode, note to note. In walking mode, a voice ignores the Form Pitch curve and follows the Melodic Probabilities. The value ranges from 0.0 (random melody) to 1.0 (walking up and down the scale). A value between 0.0 and 1.0 represents the probability of the voice being in walking mode, note to note. This makes it possible for a single voice to randomly be in either walking or random (jumpy) mode.

5.8 Pan

Pan is the MIDI stereo pan for the voice.

5.9 Follow/Leader

The Follower/Leader feature can tell a voice to follow (canon) another voice while ignoring the Form curves altogether. Set the follow button to make a voice follow another. Set the Leader to the number of the voice to follow. Since the follower voice is written after the leader has been completely finished,

5.9.1 Interval Type/Interval

The follower’s Interval Type defines whether the follower stays in the scale for the form, or if it follows on a chromatic interval. If the interval type is scalar, then the interval is the number of steps up or down the scale that the follower appears. If the interval type is chromatic, then the follower will be "Interval" number of chromatic steps away from the voice it is following.

5.9.2 Delay

The follower’s Delay creates a canon effect in that it delays the start of the following voice for the duration given as an ordinary rational number of whole notes. It can be set to 1/4 for a quarter note, or 13/17, or 4/1 or 4 for four whole notes delay and so on. It interprets 4 as four whole notes and not as a quarter note the way `textmidi` does in LAZY mode.

5.9.3 Duration Factor

The follower’s Duration Factor multiplies all of the leader’s rhythms by a factor that is given as an ordinary rational number. It can be set to 1/4 for making the follower four times faster, 4/1 or 4 to be four times slower, or other rational numbers. It interprets 4 as four whole notes and not as a quarter note the way `textmidi` does in LAZY mode.

5.9.4 Inversion

The follower can be an inversion of the leader voice, for the entire duration of the piece. Transposition interval and chromatic or scalar interval type also apply.

5.9.5 Retrograde

The follower will be a retrograde (that is, a backwards copy) of the leader voice, for the entire duration of the piece.

5.10 Random Program

The Random Program feature can make a given track (musical line) randomly change its program (or patch, or instrument) from note to note. This feature was inspired by John Cage's composition *Cheap Imitation* as rendered for orchestra, as well as Anton Webern's orchestration of Bach's *Ricercar* from *The Musical Offering*. Also note measure 97 of the Andante con moto from Beethoven's c minor symphony. Each voice has its own probability and ensemble for random programs. Random Programs apply only to lead voices, not followers. However, follower voices will copy the changing random instruments of the leads.

5.10.1 Probability

The field probability gives the probability that the next note will get a randomly-selected program. If the probability is set to 1.0, then every note will get a random instrument (but if it is the same as is in place then no PROGRAM command will be written). If the probability is .5, then on average every other note will randomize its MIDI program. If the probability is zero, then the random ensemble will have no effect at all.

5.10.2 Ensemble

The ensemble is a list General MIDI programs from which a voice can randomly choose a program. By default, `textmidicgm` uses the key number (pitch) selected for a note to find which programs are realistically available, given their respective ranges. The ranges used are seen in the program spin button near the top of the voice window. If no instrument is found that supports that pitch, then it generates a rest instead of a note event. If the command-line option *ignoreranges* is specified for `textmidicgm`, then instead of looking up instruments by key first, `textmidicgm` will just randomly select a program from the ensemble. The NoteEvent class generates a LAZY mode version of the PROGRAM message for that program prior to the current note being added to the textmidi file. Note that the melodic instruments in General MIDI (1.0) range from 1 to 112 (in a conservative assessment).

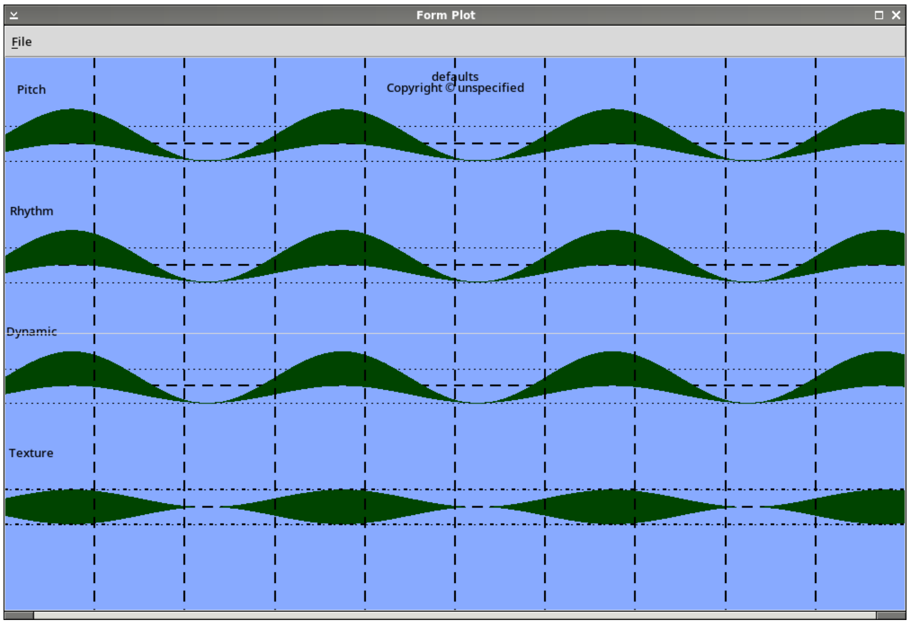
5.11 Add Button

The Add button copies the program that is visible in the Program spin button near the top of the voice window into the ensemble list. You can select different instruments to add to the ensemble by using the Program spin button to show different programs. Note that changing the voice's program in order to build a random ensemble does change the voice's instrument, but that may not matter since you are using a random ensemble in any case. However you can set the voice's program back to whatever you like, in case you might change the random program probability back to zero.

5.12 Delete Button

The Delete button removes the currently showing program name from the program list and then displays the first program in the remaining list.

6 Form Plot Window



The Form Plot window has a menu. The Form Plot Window shows the sinusoidsal curves that act to change the mean and range of pitch, rhythm (duration), dynamic (loudness) and the complement (the number of instruments playing). A vertical line is drawn at each minute of time from the beginning. The Form Plot Window menu has several items.

6.1 Open

The Open menu item (accelerator 'O') allows you to load a boost::serialization (cf. boost.org) XML archive form file. All of the values of the form file will be distributed throughout the user interface and the plot redrawn.

6.2 Save

The Save menu item (accelerator 'S') allows you to save an XML form file. All of the values of the form in the user interface of `TextMidiFormEdit.py` will be written out to the file. The XML form file is readable by `textmidicgm` using the –xmlform option.

6.3 Defaults

The *Defaults* menu item (accelerator 'F') sets all values to basic values (not randomly selected values). To generate random forms, use the *Randomize* menu item or `textmidicgm` –random filename. This does not randomize either the scale or the Random Program feature.

6.4 Randomize

The *Randomize* menu item (accelerator 'R') makes randomized selections and values in the form except the following:

- The scale is set to a full chromatic scale.
- The length is set to a half hour.
- Minimum note length is set to 0.00128 seconds.
- Ticks per quarter is set to 1440.
- The beat is set to a quarter note.
- The meter is set to 4/4.
- The tempo is set to M.M. 60 per quarter.
- No voice is set to follow.
- No voice is set with a random ensemble.

6.5 Redraw

The *Redraw* menu item (accelerator 'D') redraws the form plot; this is called for if the form curves are changed in the Musical Form window.

6.6 Save HTML

The Save HTML menu item (accelerator 'H') will save the form data to an HTML file that may be suitable for use in reports. The format of the HTML attempts to make the form data easier to see in a glance. It does not include the form plot, but that can be saved as a Postscript file. Random Program blocks are not written if the probability is zero.

6.7 Save Postscript

The Save Postscript menu item (accelerator 'P') will save a Postscript image of the Form Plot window.

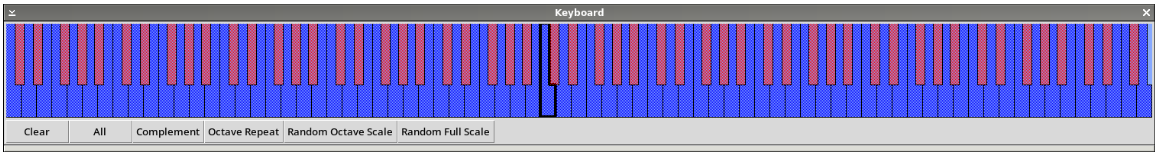
6.8 About

The About menu item (accelerator 'A') puts up a window with information about the program.

6.9 Quit

The Quit menu item (accelerator 'Q') will exit the program. The window close decoration will also do this. Any work done but unsaved is lost; you are not prompted to save work.

7 Keyboard Window



The Keyboard Window has a keyboard in which individual keys can be clicked, and also several buttons. The keyboard presents 128 keys, all of the keys supported by MIDI (C-1 to G9). Middle-C is outlined in a thicker line than other keys. Any key can be clicked to toggle whether it is to be included in a scale. When a key is included in a scale it is drawn with a stipple in blue (if a white key) or red (if a black key). Keys that are not in the scale are white if they are white keys and black if they are black keys. The keyboard will reflect selection of scale made in the Musical Form window, and the scale in the keyboard window can be copied into the form by clicking on the "From Keyboard" button in the Musical Form window. The keyboard does not affect the scale in the Form until you click "From Keyboard". Clicking on a key toggles its inclusion in the scale. If you click on a selected key, it is removed from the scale; if the key is not in the scale, then clicking on it adds it to the scale. Any arbitrary combination of the 128 keys can be clicked to create a scale. To create a scale that repeats each octave, just click the unique keys for one octave, then hit the "Octave Repeat" push button.

7.1 Clear

Clear the scale.

7.2 All

Set all of the keys, that is, the full chromatic MIDI scale.

7.3 Complement

Set the scale to select notes not in the scale and remove notes that were in the scale. For example, select a Diatonic scale in the Musical Form window; this will appear in the keyboard as C Major. Then push the Complement push button on the Keyboard window to select the missing notes instead, and the result will be the black keys only, which is a form of pentatonic scale.

7.4 Octave Repeat

Any notes set in the keyboard are repeated per octave throughout the full scale.

7.5 Random Octave Scale

Create a random scale contained within an octave. If you press Octave Repeat, this one-octave scale will fill the keyboard. Push Clear first to see the new scale.

7.6 Random Full Scale

Create a random scale across the keyboard. The scale is unlikely to repeat for each octave. Push Clear first to see the new scale.

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